



TRACK: SOFTWARE WEB DEVELOPMENT
(Front-end React)

Wethearly

Graduation Project Documentation

TEAM MEMBERS:

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Project Idea

Weatherly is a modern weather forecasting web application designed to provide users with accurate and real-time weather information for any city around the world.

The application allows users to search for locations instantly and displays detailed weather data, including current temperature, humidity, wind speed, air pressure, visibility, UV index, sunrise and sunset times, and a five-day weather forecast.

The main objective of the project is to provide a clean, responsive, and user-friendly interface that makes weather information easy to understand while delivering fast performance through API integration.

Technologies Used

To ensure high performance, clean component modularity, and smooth single-page user interactions, the platform was developed using the following technological stack:

React.js

React Router DOM

Bootstrap

Axios

OpenWeatherMap API

HTML5

CSS3

JavaScript (ES6+)

Project Planning & Management

Project Proposal & Scope: Weatherly is a modern, single-page weather forecasting web application. The primary objective is to consume live data from the OpenWeatherMap API and present it through a highly interactive, animated, and responsive user interface.

Team Roles:

- **Nada Reda Taha:** UI/UX Implementation and CSS animations.
- **Adel Abdel Halim :** API integration and data fetching logic.
- **Aya Abdel Hamid & maximus helmy:** Core React component architecture, state management, and local development environment structuring.
- **Menna Allah Shaker:** Version control management and deployment pipeline.

Timeline Tracking: The project milestones are actively aligned with the final student submission deadlines, targeting full implementation completion and final presentation readiness by mid-July 2026.

Requirements Gathering

User Stories

- As a user, I want to search for any city globally so that I can see its current weather conditions immediately.
- As a user, I want to view a 5-day forecast so that I can plan my week.
- As a user, I want to see a visual graph of temperature trends for better data comprehension.

Functional & Non-Functional Requirements

Requirement Type	Description
Functional	Search functionality, 5-day forecast display, dynamic background changes based on weather, metric/imperial unit toggle.

Requirement Type	Description
Non-Functional	Cross-device responsiveness (Mobile/Tablet/Desktop), API response time under 2 seconds, secure handling of API keys.

Testing & Quality Assurance

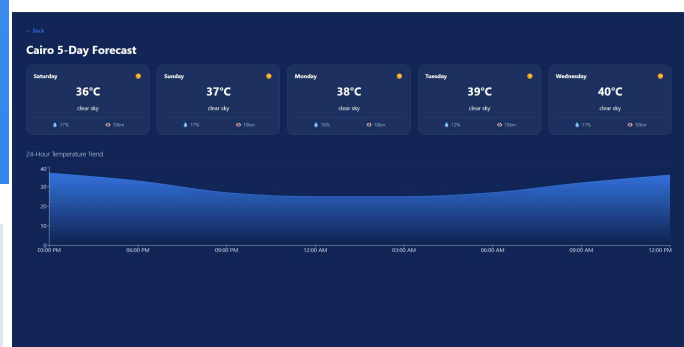
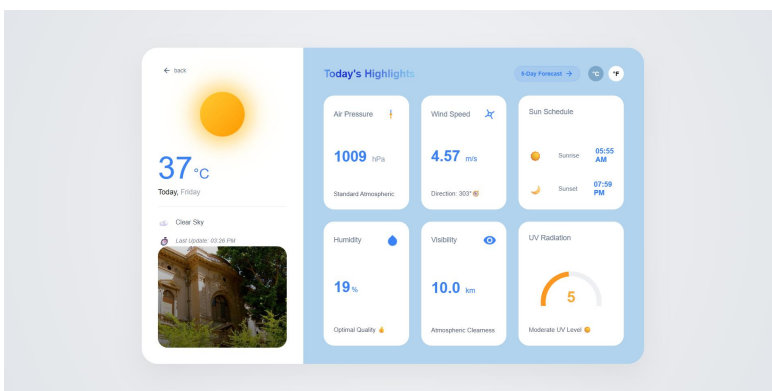
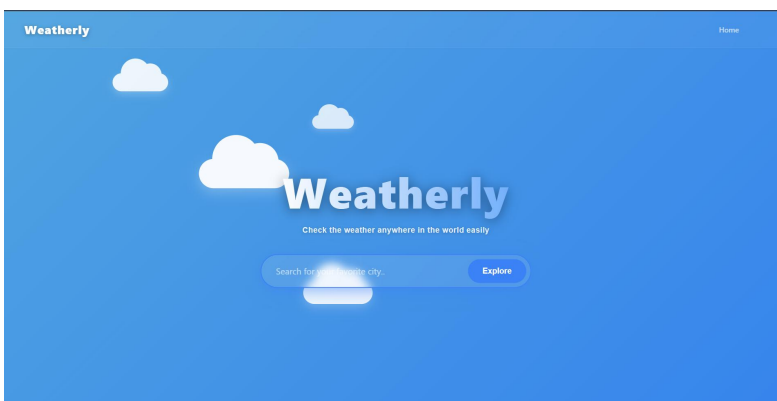
Since the project is strictly evaluated on the Front-end React track, the system design focuses entirely on client-side architecture rather than traditional database modeling.

Software Architecture

- **Component-Based UI:** The interface is divided into modular React components (e.g., SearchBar, WeatherCard, ForecastChart) to ensure reusability.
- **State Management:** React Hooks (useState, useEffect) manage API data payloads and UI loading states dynamically without page reloads.
- **Routing:** React Router DOM handles single-page navigation to maintain a seamless user experience.

UI/UX Design

The interface relies on Framer Motion for smooth component transitions. The color palette dynamically shifts using CSS3 variables based on the time of day (sunrise/sunset data) retrieved from the API.



Testing & Quality Assurance

All source code follows modern ES6+ JavaScript standards. Components are strictly modular. Axios is utilized for its promise-based structure, allowing clean async/await syntax when fetching nested JSON payloads from OpenWeatherMap.

Version control strategy (Github)

The team utilizes a standard Git workflow. Main features are developed on separate branches and merged into a central development branch via Pull Requests to avoid merge conflicts.

System Deployment & Execution (README Guide)

The following steps outline the execution guide for setting up the project environment:

1. Clone the repository

```
git clone https://github.com/YourOrganization/Weatherly.git
```

2. Navigate to the project directory

```
cd Weatherly
```

3. Install dependencies

```
npm install
```

4. Local Development Workflow

Developers can edit HTML, CSS, and React JSX files locally on their machines.

Save the files to trigger hot-reloading and verify UI changes before server deployment.

5. Start the local server

```
npm start
```

Testing & Quality Assurance

- **Component Testing:** Ensuring all UI components render correctly across different screen sizes using browser DevTools.
- **API Error Handling:** Implementing try/catch blocks to render friendly error messages (e.g., "City Not Found") when a user inputs invalid data.
- **Performance Testing:** Optimizing React re-renders to ensure smooth Framer Motion animations and fast Recharts data rendering.